

1. Circular Shift Let A be an array of n distinct integers which are sorted into increasing order and then circularly shifted by k positions to the right. For instance, $[3, 4, 8, 11, 15, 18]$ is sorted and shifted by $k = 3$ into $[11, 15, 18, 3, 4, 8]$.

Given such an array which has been shifted by an unknown k , describe a logarithmic-time algorithm that determines the value of $k \bmod n$.

2. Twisted cable We have a long data cable with n individual wires exposed on either end; each wire on the left end is connected to exactly one wire on the right end. We need to find out which wires map to which in order to transmit data efficiently, but the cable has been twisted, so this mapping is not obvious.

We can use the following operations:

- (a) connect electricity to a given wire on the left end,
- (b) disconnect electricity from a given wire on the left end and
- (c) measure electricity on a given wire on the right end.

Design an algorithm to find out which wire is connected to which while minimizing the number of these operations.

3. Weighted Median Consider n numbers $p_1, \dots, p_n$ with associated positive weights $w_1, \dots, w_n$ which satisfy that $\sum_{i=1}^n w_i = 1$. We say that p_k is a *weighted median* of the given numbers if $\sum_{p_i < p_k} w_i < \frac{1}{2}$ and $\sum_{p_i > p_k} w_i \leq \frac{1}{2}$. Describe an algorithm which finds a weighted median of given numbers in time $O(n)$. The input numbers are given in an array which is not sorted.

4. Library Say that we have a sequence of n books in alphabetical order—each with a width w_i and a height h_i —and a single library cabinet with several shelves of fixed width W but mutable height. We want to arrange the books so as to minimize the total height of the cabinet while preserving the alphabetical ordering.